

4.7 Consider the employee database of Figure 4.12. Give an SQL DDL definition of this database. Identify referential-integrity constraints that should hold, and include them in the DDL definition.

employee (ID, person_name, street, city)
works (ID, company_name, salary)
company (company_name, city)
manages (ID, manager_id)

Figure 4.12 Employee database.

-- 雇员表

```
create table employee (  
    ID varchar(10) primary key,  
    person_name varchar(50) not null,  
    street varchar(100),  
    city varchar(50)  
);
```

-- 公司表

```
create table company (  
    company_name varchar(100) primary key,  
    city varchar(50)  
);
```

-- 员工表

```
create table works (  
    ID varchar(10) primary key,  
    company_name varchar(100),  
    salary decimal(10,2)  
    foreign key(ID) reference employee(ID)  
        on delete cascade  
        on update cascade;  
    foreign key(company_name) reference company(company_name)  
        on delete cascade  
        on update cascade;  
);
```

-- 经理表

```
create table manages (  
    ID varchar(10) primary key,  
    manager_id varchar(10),  
    foreign key(ID) reference employee(ID)  
        on delete cascade  
        on update cascade;  
);
```

4.16 Write an SQL query using the university schema to find the ID of each student who has never taken a course at the university. Do this using no subqueries and no set operations (use an outer join)

```
-- 筛选出没选择选修课的学生 ID
select student.ID
from student left outer join takes
on student.ID = takes.ID
where takes.course_id is null;
```